

1.2 Applications of Propositional Logic

Translating English Sentences

- There are many reasons to translate English sentences into expressions involving propositional variables and logical connectives.
- In particular, English is very ambiguous.
- Translating sentences into compound statements removes the ambiguity.
- Once we have translated sentences from English into logical expressions, we can analyze these logical expressions to determine their truth values, we can manipulate them, and we can use rules of inference to reason about them.

Example 1: How can this English sentence be translated into a logical expression? "You can access the Internet from campus only if you are a computer science major or you are not a freshman."

Soln:

$$\begin{aligned} p &\rightarrow \text{You can access the Internet from campus} \\ q &\rightarrow \text{You are a computer science major} \\ r &\rightarrow \text{You are a freshman} \end{aligned}$$

$$p \rightarrow (q \vee \neg r)$$

Example 2: How can this English sentence be translated into a logical expression? "You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old."